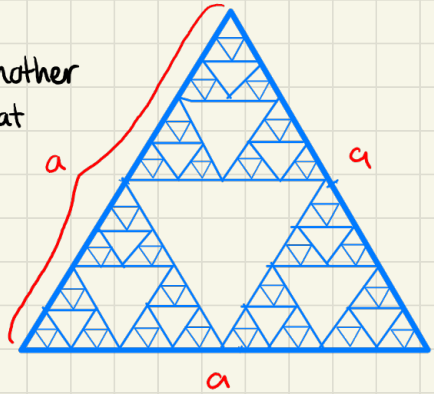
feb. 10

- Math hiring candidate teaching, Dr. Hayley Milbourne
  - Pronouns: she, her, hers
  - Ph.D in Mathematics Education
  - B.S from UCSD in scientific computation
  - M.S from University of Arizona in applied mathematics
  - Director of the Math Learning Center } at Uni. of San Diego
  - Director of the Math Placement Exam
  - Have taught at 5 different universities and colleges

- fractals
  - Geometric figures that appear the same at different scales
  - Often exhibit similar patterns at increasingly smaller scales, a property called self-similarity
  - Give us insights into fractional dimensions, which measure complexity
    - Complexity measures how thoroughly an object fills space, which is very useful in characterizing irregular natural phenomena

- Sierpiński's Triangle
  - Start with equilateral triangle, draw another w/ vertices at side midpoints, delete that area, repeat
  - What can we ask about the figure?
    - Area?
    - Perimeter?
    - Convergence?
    - Scale (changing)?
    - What points left over when put in Euclidean plane? is the set of points connected?



- Today's goal: find the area and perimeter of Sierpiński's Triangle as the process of creating it goes to infinity
  - Area: decreasing, let area = A

$$A_n = A_{n-1}\left(\frac{3}{4}\right) = A_0\left(\frac{3}{4}\right)^n = \frac{\sqrt{3}}{2}a^2\left(\frac{3}{4}\right)^n$$

$\lim_{n\to\infty} \frac{\sqrt{3}}{2}a^2\left(\frac{3}{4}\right)^n = 0$, so area converges to 0

- Perimeter: increasing

cut existing lengths in half to make new ones

Initial, $P_0 = 3a$, $P_1 = 3a + \frac{3}{2}a$, $P_2 = 3a + \frac{3}{2}a + \frac{9}{4}a$

$P_n = \underbrace{3a + \sum_{i=1}^{n} \left(\frac{3}{2}\right)^i a}$ → this diverges, keeps getting bigger, grows without bound

- So, the triangle has 0 area and ∞ perimeter

fractional dimensions

- <u>Fractal Dimension</u> - what would the dimension be for Sierpiński's Triangle?
  - Can't be 2D, wouldn't allow 0 area and ∞ perimeter
  - We define dimension in the following way: $r^{-D} = N$
    - r is the scaling factor $= 1/2$
    - D is the dimension $= ?$
    - N is the # of self-similar pieces $= 3$

ex. line, cutting in half, ex. square, cut sides in half

$\frac{1}{2}^{-D} = 2$, $D = 1$

$\frac{1}{2}^{-D} = 4$

$D = 2$

$\frac{1}{2}^{-D} = 3 \Rightarrow -D\ln(1/2) = \ln(3) \Rightarrow D = \frac{-\ln(3)}{\ln(1/2)} \approx 1.5\ldots$

Triangle is in 1.5... dimension

☆ Sierpiński's Tetrahedron lives in 2D, even though it is a solid

- Summary
  - fractals are infinitely repeating geometric shapes, many of which are self-similar
  - You can find approximate fractals in many parts of nature, particularly in the structure of cells
  - fractal dimension is defined by a relationship between the magnification factor and the number of self-similar pieces
    - This means fractals exist "between" dimensions

- Back to our class

☆ Office hrs. on friday moved, exact time to be determined

- $\mathbb{Z}/m\mathbb{Z} = \{\bar{0}, \bar{1}, \ldots, \overline{m-1}\}$

  ↳ $\{n \in \mathbb{Z} : n \equiv 1 \bmod m\}$

  - We can add & multiply in this set
    - $\bar{a} + \bar{b} = \overline{a+b}$, $\bar{a} \cdot \bar{b} = \overline{ab}$, both well-defined
  - Additive group, $(\mathbb{Z}/m\mathbb{Z}, +)$
    - set has order $|m|$ ↳ $\mathbb{Z}_m$

Ex $|\bar{2}_5|$, $\bar{2} \in (\mathbb{Z}/5\mathbb{Z}, +)$

$\bar{2}$, $\bar{2} + \bar{2} = \bar{4}$, $\bar{2} + \bar{2} + \bar{2} = \bar{6} = \bar{1}$, $\ldots$, $\underbrace{\bar{2} + \cdots + \bar{2}}_{5 \text{ times}} = \overline{10} = \bar{0}$, so $|\bar{2}_5| = 5$

- Multiplicative group, $(\mathbb{Z}/m\mathbb{Z}, \cdot)$

  ?

  - Not a group, $\bar{0}$ has no inverse
  - Instead, take set of elements of $\mathbb{Z}/m\mathbb{Z}$ with mult. inverses, which <u>is</u> a group under multiplication
    - This set has many labels: $\mathbb{Z}/m\mathbb{Z}^{\times}$, $\mathbb{Z}/m\mathbb{Z}^{*}$, $U(\mathbb{Z}/m\mathbb{Z})$, $\mathbb{Z}_m^{\times}$, $\mathbb{Z}_m^{*}$, $U(m)$

      "units" divisors of 1

      written this way in the book, kind of awful notation
    - Same as $\{\bar{a} \in \mathbb{Z}/m\mathbb{Z} : \gcd(a,m) = 1\}$
  - Bézout's Thm. says $ax + my = 1$ with $x, y \in \mathbb{Z}$ iff $\gcd(a,m) = 1$.

    $\hookrightarrow \Leftrightarrow \bar{a} \cdot \bar{x} = \bar{1}$